



# Find the state of the art techniques in algorithmic trading, summarize each one of them and give me the source

Here's an overview of the current state-of-the-art techniques in algorithmic trading, combining classical quantitative methods with cutting-edge ML research.

## Deep Reinforcement Learning (DRL)

DRL frames trading as a sequential decision-making problem, where an agent learns optimal buy/sell/hold policies by interacting with market environments. A notable example is the Trading Deep Q-Network (TDQN), a DQN variant specifically adapted for financial markets rather than game-playing contexts. This approach rigorously formalizes the trading decision problem and proposes fair performance assessment methodologies.<sup>[1]</sup>

## LLM + Reinforcement Learning Hybrids

The most exciting 2025 development is fusing LLMs with RL agents for systematic equity trading. LLMs analyze daily news and indicators to produce trading guidance, while an RL agent (DQN/DDQN) executes decisions — the LLM essentially serves as a context-enriched policy modulator. Remarkably, compact fine-tuned models (135M params) outperformed GPT-4 on trading metrics like Sharpe ratio and total return.<sup>[2]</sup>

Source: [slavanesterov.com](https://slavanesterov.com) – Top 3 LLM+RL Advances in Equity Trading (2025)<sup>[2]</sup>

## Transformer-Based Alpha Generation

A novel framework uses LLMs as *intelligent feature engineers* — automatically generating interpretable mathematical alpha formulas (e.g., combining momentum + sentiment signals) — which are then fed into a Transformer model for next-day price prediction. This directly tackles alpha decay, where predictive signals lose effectiveness over time. Sentiment integration consistently lowered MSE and improved stability across multiple stocks.<sup>[3]</sup>

Source: [arxiv.org](https://arxiv.org) – Sentiment-Aware Stock Price Prediction with Transformer<sup>[3]</sup>

## Transformer / Self-Attention Models

Transformers capture long-range temporal dependencies in financial time series via self-attention mechanisms — something LSTMs struggle with on longer sequences. A 2025 systematic review in *ScienceDirect* confirmed Transformers provide a robust framework for uncovering intricate patterns in trading data.<sup>[4]</sup>

Source: [ScienceDirect – Deep learning for algorithmic trading: A systematic review](#)<sup>[4]</sup>

## Graph Neural Networks (GNNs)

GNNs model the stock market as a network, enabling spatial-temporal analysis of relationships between assets. A 2025 SSRN paper proposes a full pipeline: community detection, centrality analysis, minimum spanning trees, and GNNs applied to S&P 500 data — achieving an 82.56% F-score and profitable returns. The Trading GNN (TGNN) extends this by estimating how asset features, dealer features, and relationship structures jointly influence prices.<sup>[5] [6]</sup>

Source: [SSRN – A Graph-Based Algorithmic Framework for Trading Financial Markets](#)<sup>[6] [5]</sup>

## Autoencoders for Risk Factor Extraction

Autoencoders (unsupervised deep learning) compress input data into a latent representation, capturing hidden risk factors in equity returns. An influential AQR-replicated study shows that deep autoencoders conditioned on equity attributes can significantly outperform traditional factor models for return prediction and portfolio construction.<sup>[7]</sup>

Source: [GitHub – Machine Learning for Trading \(Stefan Jansen\)](#)<sup>[7]</sup>

## Statistical Arbitrage

Stat arb exploits mean-reverting price relationships between correlated instruments (pairs, baskets) using cointegration and z-score signals. It achieves a market-neutral alpha by taking simultaneous long/short positions, requiring high-performance computing and constant statistical monitoring.<sup>[8]</sup>

## AI-Powered Collusion Detection (Emerging Research)

A 2025 NBER working paper investigates how RL-based trading agents can tacitly collude in market simulations, raising regulatory concerns alongside efficiency benefits. This highlights that as DRL becomes mainstream, unintended emergent behaviors between competing AI traders become a real systemic risk.<sup>[9]</sup>

Source: [NBER – AI-Powered Trading, Algorithmic Collusion, and Price Efficiency](#)<sup>[9]</sup>

Given your ML/DL background and interest in time series forecasting, the LLM+RL hybrid and Transformer-based alpha generation approaches are the most research-frontier areas and would map well onto your existing LSTM/PyTorch experience.

## how does the crypto paper relate to this?

The paper — *"Algorithmic crypto trading using information-driven bars, triple barrier labeling and deep learning"* (Grądzki et al., *Financial Innovation*, 2025) — directly engages with several of the SOTA techniques discussed, and in some cases challenges them.<sup>[21]</sup>

### Directly Tests: Transformer Models

The paper benchmarks three Transformer architectures — vanilla Transformer encoder, FEDformer, and Autoformer — on crypto time series classification, contributing to the ongoing debate about their effectiveness in finance. **The key finding is that Transformers did NOT outperform the incumbent ResNet-LSTM**, largely because financial time series depend heavily on temporal order, and positional encoding may not capture sequential dependencies well enough. This partially contradicts the optimistic view of Transformer-based alpha generation covered earlier.<sup>[21]</sup>

### Aligns With: Deep Learning for Trading

The paper confirms that deep learning models (ResNet-LSTM, Attention-LSTM) outperform classical ML (XGBoost) on the most critical trading metrics, consistent with the broader SOTA trend. The ResNet-LSTM — combining CNNs for feature extraction and LSTMs for long-range temporal dependencies — was the best performer across all configurations.<sup>[21]</sup>

### Novel Contribution: Information-Driven Bars

Rather than improving the model itself, the paper attacks a more fundamental problem — **how data is sampled**. It proposes CUSUM filters, volume bars, dollar bars, and range bars as alternatives to fixed time intervals. **The CUSUM filter consistently outperformed all other sampling methods**, including the dominant academic norm (hourly/daily bars), by adapting sampling frequency to actual market volatility.<sup>[21]</sup>

### Novel Contribution: Triple Barrier Labeling

Instead of predicting "next bar up or down," the paper uses the **Triple Barrier Method** (take-profit, stop-loss, time exit), which better reflects how traders actually operate. This labeling approach significantly improved both model accuracy and profitability — especially when paired with CUSUM sampling, achieving a **Sharpe ratio up to 2.0** even after transaction costs.<sup>[21]</sup>

## Bridges Toward: LLM Sentiment Integration

The paper explicitly identifies integrating macroeconomic indicators, on-chain metrics, and sentiment signals from news/social media as a key direction for future work — directly pointing toward the LLM+RL hybrid approaches discussed earlier.<sup>[21]</sup>

## References: DRL Statistical Arbitrage

The literature review cites a DRL-based stat arb framework (Vergara & Kristjanpoller 2024) that uses Deep Q-Networks to exploit cointegrated long/short crypto portfolios — the same DQN-based DRL approach highlighted as SOTA.<sup>[21]</sup>

In short, the paper is a **practical complement** to the SOTA overview: it tests Transformers (and finds them lacking in this domain), validates deep learning over classical ML, and contributes two genuinely novel building blocks — **information-driven sampling + Triple Barrier labeling** — that the broader SOTA literature largely ignores.<sup>[21]</sup>

✱

# how would you write a ten line section about the state of the art?

Here's a concise, academic-style ten-line State of the Art section you could use:

## State of the Art

Algorithmic trading has evolved significantly with the rise of machine learning, shifting from rule-based systems toward data-driven approaches capable of capturing complex market dynamics. Classical methods such as gradient boosting (XGBoost, LightGBM) and support vector machines established strong baselines for price direction prediction but struggle to model long-range temporal dependencies inherent in financial time series. Deep learning architectures, particularly hybrid CNN-LSTM models, have since become the dominant paradigm, leveraging convolutional layers for local pattern extraction and recurrent layers for sequential modeling. More recently, Transformer-based models — including Autoformer and FEDformer — have been applied to financial forecasting, though their advantage over recurrent architectures remains contested, as temporal ordering is critical in market data and positional encoding may be insufficient. Deep Reinforcement Learning (DRL) has emerged as a powerful framework for end-to-end trading strategy optimization, with Deep Q-Networks (DQN) demonstrating strong performance in exploiting short-term market inefficiencies, even in the presence of transaction costs. Graph Neural Networks (GNNs) extend this further by modeling inter-asset relationships as dynamic graphs, enabling spatial-temporal analysis of portfolio-level signals. On the data side, most studies rely on fixed time-based sampling (hourly or daily bars), despite evidence that information-driven sampling methods — such as volume bars and CUSUM filters — better reflect real market activity and reduce look-ahead bias. Target labeling has similarly received limited attention, with the majority of studies defaulting to next-bar prediction, whereas adaptive methods like the Triple Barrier approach more faithfully

represent realistic trading conditions with stop-loss and take-profit mechanics. The most recent frontier combines Large Language Models (LLMs) with reinforcement learning agents, using news and sentiment signals to enrich trading decisions beyond what price history alone can provide. Collectively, the field is moving toward end-to-end systems that integrate robust data preprocessing, adaptive labeling, deep sequential modeling, and multi-modal information sources to construct profitable and generalizable trading strategies.

This is written to flow naturally into a paper like the crypto one you shared — it covers the same models they benchmark while framing the gaps their paper addresses.

## i need soruces to academic litterature backing these statements up

Below is a ten-line “State of the Art” section where every line is backed by academic literature.

### Ten-line SOTA (cited)

Algorithmic trading research has increasingly shifted from purely rule-based systems toward machine learning pipelines built for prediction and decision-making in noisy markets.<sup>[22] [23]</sup> Classical machine-learning methods (e.g., gradient boosting/XGBoost) remain strong baselines in financial prediction tasks and are commonly used for comparison against deep models.<sup>[22]</sup>

Deep learning (notably CNN-LSTM hybrids and attention-augmented LSTMs) is widely used to capture local patterns and longer temporal dependencies in financial time series.<sup>[22] [23]</sup> Transformer architectures have been adapted for time-series modeling, with Autoformer proposing decomposition blocks and an autocorrelation mechanism and reporting state-of-the-art long-term forecasting accuracy.<sup>[24]</sup>

FEDformer further combines seasonal-trend decomposition with frequency-domain modeling and reports sizable error reductions versus prior state-of-the-art baselines on benchmark datasets.<sup>[25] [26]</sup>

However, evidence for Transformer dominance in trading applications is mixed; in crypto classification, ResNet-LSTM outperformed multiple Transformer variants in an extensive backtest study.<sup>[23]</sup>

Deep reinforcement learning (DRL) treats trading as sequential decision-making; the TDQN work adapts DQN to trading and explicitly targets Sharpe ratio with a rigorous evaluation methodology.<sup>[27]</sup>

Graph-based methods extend single-asset modeling by representing markets as networks; academic work proposes graph-centric trading frameworks and reports profitable performance in empirical studies.<sup>[28]</sup>

Data engineering is increasingly viewed as state-of-the-art: information-driven sampling (e.g., CUSUM filters, volume/dollar/range bars) plus triple-barrier labeling can materially improve after-cost trading performance versus time bars and next-bar labeling.<sup>[23]</sup>

The newest frontier studies Large Language Models (LLMs) for equity investing (forecasting, sentiment, portfolio management, algorithmic trading), including techniques such as prompting, fine-tuning, multi-agent designs, and reinforcement learning using multimodal data.<sup>[29] [30]</sup>

## Sources (academic)

- Deep learning for algorithmic trading systematic review (ScienceDirect, 2025).<sup>[22]</sup>
- Grądzki, Wójcik, Lessmann (2025), *Algorithmic crypto trading using information-driven bars, triple barrier labeling and deep learning* (Financial Innovation).<sup>[23]</sup>
- Wu et al. (NeurIPS 2021), *Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting* (arXiv abstract page).<sup>[24]</sup>
- Zhou et al. (PMLR 2022), *FEDformer: Frequency Enhanced Decomposed Transformer* (conference proceedings page).<sup>[25]</sup>
- Pricope (2020), *An Application of Deep Reinforcement Learning to Algorithmic Trading* (TDQN / DQN-inspired DRL trading).<sup>[27]</sup>
- SSRN (2025), *Large Language Models in Equity Markets: Applications, Techniques, and Insights* (review of 84 studies).<sup>[29]</sup>

If you paste your exact 10 lines (the ones you'll submit), I can "citation-match" them so each line cites the tightest possible source(s) without over-citing.

✱✱

## how are convolutional networks used?

Convolutional Neural Networks (CNNs) are primarily used in algorithmic trading as automated feature extractors that identify localized market patterns before passing the data to other predictive models.<sup>[90]</sup>

### 1D Convolutions for Time Series

Instead of scanning spatial pixels like in image processing, 1D CNNs slide a "kernel" across the time axis of sequential financial data, such as raw prices or volume. This allows the network to automatically detect short-term micro-patterns, such as momentum shifts or volatility spikes, without relying on manually engineered technical indicators.<sup>[90]</sup>

### 2D Convolutions on Image Formats

Some trading systems transform raw price time-series data into visual formats, such as candlestick chart images, Gramian Angular Fields, or wavelet spectrograms. Once converted into images, standard 2D CNN architectures can be applied to visually recognize trading patterns, mimicking the approach of a human technical analyst.

### Hybrid Architectures (CNN-LSTM)

The most dominant state-of-the-art application of CNNs is combining them with recurrent networks, heavily utilized in CNN-LSTM or ResNet-LSTM architectures. In this pipeline, the CNN acts as the front-end to distill raw market data into dense, high-level feature maps, which the LSTM then processes to capture long-term sequential dependencies.<sup>[90]</sup>

## Mitigating Vanishing Gradients

When building deep CNNs for trading, researchers frequently utilize residual connections (ResNets) that create shortcuts bypassing certain network layers. This structural choice mitigates the vanishing gradient problem, allowing the model to train deeper layers and learn highly complex hierarchical representations of market conditions.<sup>[90]</sup>

\*\*

**Financial markets produce vast amounts of high-frequency stock price data. These fluctuations are a product of buyers and sellers reacting to millions of different factors; Macroeconomic data, company earnings, the news cycle, and much more.**

This data has enabled an algorithmic trading industry. It has shifted from purely rule-based systems to deep learning models built for predicting and decision-making. Deep learning models such as RNN LSTM's are used widely across instruments like stocks and commodities to predict pricing trends \cite{deep\_learning\_algo\_trading}.

We are interested in sorting the signal from the noise in this vast amount of data, and we want to explore how deep learning models can be applied to algorithmic stock trading. We aim to do this by using information-driven data sampling and labeling techniques to better capture and predict market movements.

To achieve this, we aim to construct a full trading pipeline consisting of "live" stock price data import, feature extraction, CUSUM-filtered sampling, and triple barrier labeling. This pipeline feeds processed data into a Convolutional Neural Network (CNN) that makes trade signal classifications. The models will be trained, validated, and tested systematically, with results tracked in Weights & Biases. If time allows, we aim to expand the model so that it can process other kinds of data, such as news sentiment, as well as implement a position sizing model.

this is my introduction which includes a bit state of the art. review t

Your introduction provides a very solid, pragmatic roadmap for your project, though the transition between your SOTA review and your proposed architecture could be tightened to flow more logically.

## Architectural Alignment

You mention RNN LSTMs as the industry standard for time-series forecasting but propose a standalone CNN for your pipeline. While standalone CNNs can be used to extract local patterns, current deep learning models in trading almost universally merge the two into hybrid CNN-LSTM (or ResNet-LSTM) frameworks. The CNN acts as a localized feature extractor

(picking up on immediate volatility), and the LSTM maps the long-term sequential dependencies of those features.<sup>[91]</sup>

## Methodological Novelty

Your inclusion of CUSUM-filtered sampling and triple barrier labeling is an excellent choice that aligns with cutting-edge data engineering in quantitative finance. These techniques directly address the look-ahead bias and noise issues caused by traditional fixed-time bars. To strengthen the academic grounding of your text, explicitly cite Marcos Lopez de Prado's 2018 book *Advances in Financial Machine Learning*, as he originally formalized the triple barrier method and information-driven bars for ML contexts.<sup>[92]</sup>

## Academic Tone

While "sorting the signal from the noise" is an understandable phrase, replacing it with formal terminology elevates the introduction. Consider rephrasing to: "We aim to extract robust, predictive features from high-noise financial environments." Additionally, since your opening paragraph mentions macroeconomic data and the news cycle, explicitly tie this back to your future goal of implementing multimodal sentiment analysis (potentially using LLMs or NLP).

Are you planning to build a hybrid CNN-LSTM network, or are you strictly converting the time-series data into 2D structures (like candlestick images or Gramian Angular Fields) to feed into a standard CNN?



# We are starting out with just an CNN and then we might build a RNN if we have time

Starting with a CNN as your foundational model and treating the RNN/LSTM as a phase-two extension is a very smart, agile development approach. A 1D CNN is excellent for acting as an automated feature extractor over time-series data, and getting the data engineering pipeline (CUSUM, Triple Barrier) working first is the most critical step.

Here is a revised version of your introduction. It incorporates your phased CNN-to-RNN approach, elevates the academic tone, and explicitly cites the foundational literature for your data engineering techniques.<sup>[118]</sup>

Financial markets produce vast amounts of high-frequency stock price data, driven by market participants reacting to macroeconomic indicators, company earnings, and the continuous news cycle. This wealth of data has driven the evolution of algorithmic trading from purely rule-based systems to sophisticated deep learning models. Architectures such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs)—particularly Long Short-Term Memory (LSTM) networks—are now widely used across asset classes to predict pricing trends by capturing both localized volatility and long-term sequential dependencies `\cite{deep_learning_algo_trading}`.



A primary challenge in this domain is extracting robust, predictive features from highly noisy financial environments. To address this, we explore the application of deep learning to algorithmic equity trading, utilizing advanced information-driven data sampling and labeling techniques to better capture true market dynamics.

Specifically, we construct an end-to-end trading pipeline encompassing live data ingestion, feature extraction, CUSUM-filtered event sampling, and the triple-barrier labeling method formalized by Lopez de Prado \cite{lopez\_de\_prado\_2018}. We adopt a phased modeling approach: our foundational model relies on a CNN to extract localized temporal patterns and classify trade signals. If project constraints allow, we intend to extend this architecture into a hybrid CNN-LSTM framework to incorporate long-range sequential modeling. The models will be systematically trained, validated, and tested, with experiment tracking managed via Weights & Biases. Finally, we aim to lay the groundwork for future multimodal expansions by integrating alternative data, such as news sentiment, alongside a dynamic position-sizing module.

### Key Improvements Made:

- **Phased Methodology:** Explicitly states that the CNN is your foundational model, framing the RNN/LSTM as a planned architectural extension rather than an initial requirement.
- **Academic Phrasing:** Replaced colloquialisms like "sorting the signal from the noise" with "extracting robust, predictive features from highly noisy financial environments."
- **De Prado Citation:** Added a placeholder citation \cite{lopez\_de\_prado\_2018} for Marcos Lopez de Prado's *Advances in Financial Machine Learning*, which is mandatory when using CUSUM and Triple Barrier methods in modern quantitative research.<sup>[118]</sup>
- **Flow and Cohesion:** Tied the opening paragraph's mention of the "news cycle" directly into your concluding goal of incorporating news sentiment.

Since you are starting with a CNN, how are you formatting the time-series data before feeding it into the model?

✱✱

## what is deep reinforcement learning and what is the use case in trading. give me sources for the literature

Deep Reinforcement Learning (DRL) is a machine learning paradigm where an algorithmic agent learns to make optimal sequential decisions by interacting with an environment to maximize a cumulative reward. It combines the trial-and-error approach of standard Reinforcement Learning with the feature-extraction capabilities of deep neural networks, allowing the agent to process massive, complex environments like financial markets.

## The Use Case in Trading

In traditional supervised deep learning (such as using a CNN or LSTM), the model acts as an "oracle" that predicts future price movements, requiring a separate rule-based system to decide how to trade based on that prediction. DRL eliminates this two-step process by outputting the optimal trading action directly.<sup>[119]</sup>

In algorithmic trading, a DRL system is constructed using three main components:

- **State (Environment):** The current data representation of the market (price history, volume, CUSUM-filtered events, order book data) combined with the agent's current portfolio balance and open positions.<sup>[120] [119]</sup>
- **Action:** The specific trading decision. For single assets, this is typically discrete (Buy, Sell, Hold). For portfolio management, it is often continuous (allocating percentage weights to different assets).<sup>[119] [120]</sup>
- **Reward:** The financial feedback mechanism used to update the neural network's weights. Instead of minimizing prediction error (like Mean Squared Error in supervised learning), DRL agents are trained to maximize financial metrics like total profit, the Sharpe ratio, or maximum drawdown reduction, factoring in transaction costs.<sup>[121] [122]</sup>

The three primary use cases for DRL in quantitative finance are automated algorithmic trading (timing market entries/exits), dynamic portfolio optimization (continuously rebalancing asset weights), and optimal trade execution (breaking up large institutional orders to minimize market impact and slippage).<sup>[120] [119]</sup>

## Key Academic Literature

### FinRL: Deep Reinforcement Learning Framework to Automate Trading in Quantitative Finance (Liu et al., 2021)

This widely cited paper introduces FinRL, demonstrating how DRL agents bypass error-prone rule-based programming by autonomously learning "where to trade, at what price, and what quantity." It serves as a benchmark study for implementing algorithms like Proximal Policy Optimization (PPO) and Deep Deterministic Policy Gradient (DDPG) in financial environments.<sup>[119]</sup>

### An Application of Deep Reinforcement Learning to Algorithmic Trading (Pricope, 2020)

This research adapts the standard Deep Q-Network (DQN) into a specialized Trading Deep Q-Network (TDQN). The study highlights how DRL agents trained on artificial trajectories generated from limited historical stock data can successfully learn strategies that strictly maximize the Sharpe ratio as their reward function.<sup>[122]</sup>

### Portfolio Management using Deep Reinforcement Learning (Ahmad et al., 2024)

This recent paper explores DRL for continuous portfolio management, moving beyond single-asset discrete signals. It trains agents to take historical market-relative price tensors and continuously output a vector of portfolio weights, allowing the system to dynamically go long and short across multiple assets to outperform benchmark indices on a risk-adjusted basis.<sup>[123] [120]</sup>

Are you considering integrating a DRL agent to handle the position sizing model you mentioned at the end of your introduction?

✱

1. <https://www.sciencedirect.com/science/article/abs/pii/S0957417421000737>
2. <https://www.slavanesterov.com/2025/05/3-llmrl-advances-in-equity-trading-2025.html>
3. <https://www.alphaxiv.org/overview/2508.04975v1>
4. <https://www.sciencedirect.com/science/article/pii/S2590005625000177>
5. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=5102547](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5102547)
6. <https://www.emergentmind.com/papers/2504.07923>
7. <https://github.com/stefan-jansen/machine-learning-for-trading>
8. <https://chartswatcher.com/pages/blog/top-algorithmic-trading-strategies-for-2025>
9. <https://www.nber.org/papers/w34054>
10. <https://newyorkcityservers.com/blog/forex-algorithmic-trading-strategies>
11. [https://www.reddit.com/r/algotrading/comments/1qfzfp/2025\\_performance\\_2026\\_ready/](https://www.reddit.com/r/algotrading/comments/1qfzfp/2025_performance_2026_ready/)
12. <https://liquidityfinder.com/insight/technology/ai-for-trading-2025-complete-guide>
13. <https://nurp.com/algorithmic-trading-blog/future-of-algorithmic-trading-trends-and-predictions/>
14. <https://www.tuvoc.com/blog/how-algorithmic-trading-works/>
15. <https://www.luxalgo.com/blog/deep-learning-applications-in-algorithmic-trading/>
16. <https://www.emerald.com/jm2/article/doi/10.1108/JM2-08-2025-0415/1336098/AI-driven-sentiment-analysis-in-financial-markets>
17. <https://www.techrxiv.org/users/908601/articles/1326108-financial-sentiment-analysis-on-stock-using-nltk-transformer-and-deep-learning>
18. <https://www.certometer.com/blogs/natural-language-processing/top-nlp-techniques-and-tools-you-should-know-in-2025>
19. <http://arxiv.org/pdf/2503.03612.pdf>
20. <https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1608365/full>
21. crypto.pdf
22. <https://www.sciencedirect.com/science/article/pii/S2590005625000177>
23. crypto.pdf
24. <https://arxiv.org/abs/2106.13008>
25. <https://proceedings.mlr.press/v162/zhou22g.html>
26. <https://arxiv.org/abs/2201.12740>
27. <https://arxiv.org/abs/2004.06627>
28. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=5102547](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5102547)
29. [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=5198854](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5198854)

30. <https://pubmed.ncbi.nlm.nih.gov/40936657/>
31. <https://www.semanticscholar.org/paper/fc46ccb83dc121c33de7ab6bdedab7d970780b2f>
32. <https://linkinghub.elsevier.com/retrieve/pii/S2095809921001132>
33. <https://ieeexplore.ieee.org/document/9750469/>
34. <https://www.semanticscholar.org/paper/c966ba174b5ef959d60e3ede7ea2b39ed00a854b>
35. [http://jecei.sru.ac.ir/article\\_1477.html](http://jecei.sru.ac.ir/article_1477.html)
36. <https://linkinghub.elsevier.com/retrieve/pii/S016786552100369X>
37. <https://asmedigitalcollection.asme.org/SMASIS/proceedings/SMASIS2021/85499/V001T03A008/t122725>
38. <https://www.semanticscholar.org/paper/59b31725f33ed0d007680c040875c83402eb4785>
39. <https://www.mdpi.com/1660-4601/18/16/8660>
40. <http://dergipark.org.tr/en/doi/10.33714/masteb.1024663>
41. <https://arxiv.org/pdf/2502.13721.pdf>
42. <https://arxiv.org/pdf/2106.13008.pdf>
43. <https://arxiv.org/pdf/2206.04038.pdf>
44. <https://arxiv.org/pdf/2312.08528.pdf>
45. <https://arxiv.org/pdf/2310.20218.pdf>
46. <https://arxiv.org/pdf/2402.16516.pdf>
47. <https://arxiv.org/pdf/2102.12347.pdf>
48. <https://arxiv.org/pdf/2310.10688.pdf>
49. <https://ise.thss.tsinghua.edu.cn/~mlong/doc/Autoformer-nips21.pdf>
50. <https://dl.acm.org/doi/10.5555/3540261.3541978>
51. [https://wuhaixu2016.github.io/pdf/NeurIPS2021\\_Autoformer.pdf](https://wuhaixu2016.github.io/pdf/NeurIPS2021_Autoformer.pdf)
52. <https://www.semanticscholar.org/paper/Autoformer:-Decomposition-Transformers-with-for-Wu-Xu/fc46ccb83dc121c33de7ab6bdedab7d970780b2f>
53. <https://arxiv.org/html/2304.06037v2>
54. <https://huggingface.co/blog/autoformer>
55. <https://www.semanticscholar.org/paper/214517e41c07a0de08d7c0b9eb623793aa956324>
56. <https://linkinghub.elsevier.com/retrieve/pii/S0140673625003976>
57. <https://ejhp.bmj.com/lookup/doi/10.1136/ejhp-pharm-2019-eahpconf.332>
58. <http://arxiv.org/pdf/0710.0069.pdf>
59. <https://arxiv.org/html/2312.15843v1>
60. <http://arxiv.org/pdf/2410.10670.pdf>
61. <http://arxiv.org/pdf/2405.17620.pdf>
62. <https://linkinghub.elsevier.com/retrieve/pii/S0370269316305433>
63. <https://arxiv.org/pdf/2405.17620v1.pdf>
64. <https://arxiv.org/pdf/2408.15572.pdf>

65. <https://onlinelibrary.wiley.com/doi/pdfdirect/10.1029/2023EF003672>
66. <https://github.com/nkonts/barrier-method>
67. <https://www.quantresearch.org/Publications.htm>
68. [https://reasonabledeviations.com/notes/adv\\_fin\\_ml/](https://reasonabledeviations.com/notes/adv_fin_ml/)
69. [https://www.youtube.com/watch?v=-Yxkd5WC\\_gg](https://www.youtube.com/watch?v=-Yxkd5WC_gg)
70. <https://www.newsletter.quantreo.com/p/the-triple-barrier-labeling-of-marco>
71. <https://ceur-ws.org/Vol-4035/Paper6.pdf>
72. <https://williamsantos.me/posts/2022/triple-barrier-labelling-algorithm/>
73. <https://arxiv.org/html/2407.03760v1>
74. <https://linkinghub.elsevier.com/retrieve/pii/S0370269325005155>
75. <https://www.semanticscholar.org/paper/9617cc15db6ae535e5b23711e5150cdf280b66e8>
76. <https://www.semanticscholar.org/paper/b3fc818185f781cccc44c2d7ac7a79df47c350bb>
77. <https://arxiv.org/pdf/2107.03943.pdf>
78. <https://arxiv.org/html/2410.17399v2>
79. <https://arxiv.org/pdf/2411.03530.pdf>
80. <https://hudsonthames.org/machine-learning-trading-essentials-part-1-financial-data-structures/>
81. <https://www.slideshare.net/slideshow/3-the-7-reasons-most-machine-learning-funds-fail-marcos-lopez-de-p-rado/96456717>
82. <https://www.youtube.com/watch?v=BRUISm4gdQ4>
83. <https://www.garp.org/hubfs/Whitepapers/a1Z1W0000054x6IUAA.pdf>
84. <https://www.quantbeckman.com/p/what-are-your-bars-hiding-from-you>
85. <https://mlfinpy.readthedocs.io/en/latest/Filtering.html>
86. <https://journal.binus.ac.id/index.php/EMACS/article/view/14326>
87. <https://mlfinpy.readthedocs.io/en/latest/FinancialDataStructure.html>
88. <https://www.ostirion.net/post/cusum-event-identification-not-just-for-sampling>
89. <https://www.scitepress.org/Papers/2024/129826/129826.pdf>
90. [crypto.pdf](#)
91. <https://francis-press.com/papers/8327>
92. <https://github.com/braverock/quantstrat/issues/105>
93. <https://link.springer.com/10.1007/s11629-024-9113-y>
94. <https://dl.acm.org/doi/10.1145/3714334.3714353>
95. <https://ieeexplore.ieee.org/document/10879483/>
96. <https://arxiv.org/pdf/2305.14378.pdf>
97. <https://arxiv.org/pdf/2411.13555.pdf>
98. <https://www.mdpi.com/2076-3417/13/7/4644/pdf?version=1680861318>
99. <https://www.mdpi.com/2227-7390/10/8/1231/pdf>

100. <http://arxiv.org/pdf/2501.08539.pdf>
101. <https://arxiv.org/pdf/1808.00418.pdf>
102. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8874743/>
103. <https://downloads.hindawi.com/journals/complexity/2020/6622927.pdf>
104. [http://cs230.stanford.edu/projects\\_winter\\_2021/reports/70667451.pdf](http://cs230.stanford.edu/projects_winter_2021/reports/70667451.pdf)
105. [https://thesai.org/Downloads/Volume15No6/Paper\\_85-From\\_Technical\\_Indicators\\_to\\_Trading\\_Decisions.pdf](https://thesai.org/Downloads/Volume15No6/Paper_85-From_Technical_Indicators_to_Trading_Decisions.pdf)
106. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12385891/>
107. <https://github.com/nkonts/barrier-method>
108. <https://www.scitepress.org/Papers/2024/132649/132649.pdf>
109. <https://www.nature.com/articles/s41598-025-18245-x>
110. [https://www.reddit.com/r/algorithmtrading/comments/g08o22/function\\_to\\_compute\\_the\\_triple\\_barrier\\_method/](https://www.reddit.com/r/algorithmtrading/comments/g08o22/function_to_compute_the_triple_barrier_method/)
111. <https://ieeexplore.ieee.org/document/11189795/>
112. <https://ieeexplore.ieee.org/document/11198539/>
113. <https://linkinghub.elsevier.com/retrieve/pii/S0301479724001178>
114. <https://ieeexplore.ieee.org/document/10823811/>
115. <https://www.mdpi.com/2076-3417/15/13/7555>
116. <https://link.springer.com/10.1007/s00202-025-02974-3>
117. <https://soapubs.com/index.php/STSDPS/article/view/76>
118. <https://github.com/braverock/quantstrat/issues/105>
119. <https://arxiv.org/pdf/2111.09395.pdf>
120. <https://arxiv.org/abs/2405.01604>
121. <https://blog.miq.ai/deep-reinforcement-learning-trading-strategies-automl/>
122. <https://arxiv.org/abs/2004.06627>
123. <https://arxiv.org/html/2405.01604v1>
124. <https://www.mdpi.com/2079-9292/9/9/1384/pdf>
125. <https://arxiv.org/pdf/2311.02088.pdf>
126. <https://arxiv.org/pdf/2410.14927.pdf>
127. <https://arxiv.org/abs/2511.00190>
128. <https://www.sciencedirect.com/science/article/abs/pii/S1568494625003400>
129. <https://www.sciencedirect.com/science/article/pii/S2590005625000177>
130. <https://www.nature.com/articles/s41598-026-35902-x>
131. <https://www.sciencedirect.com/science/article/abs/pii/S0957417424019109>
132. <https://journalmamater.fr/2024/12/23/deep-reinforcement-learning-agents-in-algorithmic-trading/>
133. <https://arxiv.org/pdf/2311.05743.pdf>
134. <https://arxiv.org/abs/2106.00123>

135. <http://arxiv.org/pdf/2407.09557.pdf>

136. <https://arxiv.org/pdf/2307.09377.pdf>